

WEEK - 2 TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Intern)	Karthik Raman Keerangudi Kalyanaraman
Program / Batch	Cybersecurity Internship B083
Date	17 September 2026
Modules completed	W2-PM4 (theHarvester Footprinting — elective) W2-PM5 (Zenmap Network Scanning — essential)
Target(s)	1. microsoft.com (public footprinting exercise, OSINT sources only) 2. My own local VirtualBox / Kali lab network (10.0.0.0/24)
Permission / Authorization	Yes — public-source OSINT only against microsoft.com; own lab network for scanning
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery

1. Liability Disclaimer

I have performed these activities only on systems and data sources for which access is authorized: publicly available OSINT information for the microsoft.com domain (no active or intrusive scanning of that domain was performed), and my own self-owned local VirtualBox/Kali lab network for the scanning exercise. All activities are for education and training purposes only, completed as part of the Week 2 Cybersecurity & Ethical Hacking internship curriculum at Networkwalks. Nothing in this report should be used to access, scan, or test any system without proper authorization. Every action taken is my own responsibility. Misuse of these techniques can lead to criminal charges, heavy fines, loss of employment and a permanent record. In most countries, unauthorized access to computer systems is a crime even when no damage occurs.

2. Introduction

This report documents two Week 2 practical modules completed as part of my Cybersecurity & Ethical Hacking internship at Networkwalks: W2-PM4 (theHarvester-based footprinting, elective) and W2-PM5 (Zenmap-based network scanning, essential). Modules PM1–PM3 were not selected for this submission. Together these two exercises illustrate the first two phases of a penetration test: gathering publicly available information about a target domain (reconnaissance/footprinting), and discovering live hosts on a network (scanning).

For PM4, theHarvester was run against the microsoft.com domain using only passive, publicly available OSINT sources - the exercise was conducted using theHarvester's available public/OSINT sources and did not involve active scanning or exploitation of Microsoft infrastructure. For PM5, Zenmap/Nmap was used to scan my own local lab subnet (10.0.0.0/24) to identify live hosts and their MAC addresses, and to generate a network topology diagram. Every finding below is taken directly from the command output, text logs, screenshots and topology file captured during these two exercises.

3. Tools Used

Tool	Purpose
theHarvester 4.10.1	OSINT footprinting tool used to gather hosts, emails, IPs and ASNs for a target domain from public sources.
Kali Linux	Operating system used to run theHarvester and native networking commands (ip neigh, etc.).
Zenmap (Nmap GUI) 7.99	GUI front-end for Nmap; used to run ping scans, view results and generate a network topology diagram.
Nmap (nmap -sn)	Command-line host discovery via ICMP/ARP ping scan of the local subnet.
ip neigh	Linux command used to inspect the ARP/neighbour cache and confirm the gateway's MAC address.

4. Activities Performed

4.1 W2-PM4 — theHarvester Footprinting (elective)

The target for this module was the microsoft.com domain, used strictly as an educational OSINT exercise against public sources — no direct scanning of Microsoft's infrastructure was performed. Two separate theHarvester runs were carried out.

Exercise 1 — single source (Baidu):

```
theHarvester -d microsoft.com -l 1000 -b baidu
```

This run returned no IP addresses from Baidu, 2 email addresses, and 10 hosts (subdomains) associated with microsoft.com, including learn.microsoft.com, support.microsoft.com and news.microsoft.com.

Exercise 2 — all available sources:

```
theHarvester -d microsoft.com -l 50 -b all
```

Most third-party sources (Shodan, VirusTotal, Hunter, Censys, SecurityTrails, etc.) returned "missing API key" errors, since no paid API keys were configured — this is expected in a lab environment and only the free/unauthenticated sources actually returned data. From the sources that did return results (Baidu, CRTsh, DuckDuckGo, Hackertarget, Rapiddns, Subdomaincenter, Urlscan, Yahoo, Waybackarchive, DNS fallback pattern search, and others), theHarvester reported:

- 8 ASNs (AS13335, AS133618, AS16625, AS20940, AS24940, AS40034, AS8070, AS8075)
- 8 "interesting" URLs (public support/learn pages, e.g. support.microsoft.com and learn.microsoft.com links)
- 153 IP addresses associated with resolved hostnames
- 3 email addresses (dotnet-docker-bot@microsoft.com, opencode@microsoft.com, secure@microsoft.com)
- 9,969 host/subdomain names — almost entirely internal-style Microsoft subdomains surfaced by public certificate-transparency and search-engine sources (e.g. *.microsoft.com, *.redmond.corp.microsoft.com, *.azure.microsoft.com); this is a large but entirely passive/OSINT result set, not a list of systems that were probed or accessed
- 0 LinkedIn users/links found

No people, credentials, or non-public information were discovered in either run — all results came from indexed public sources reachable by theHarvester's free modules.

4.2 W2-PM5 — Zenmap Network Scanning (essential)

The target for this module was my own local VirtualBox/Kali lab network. The lab subnet is 10.0.0.0/24, with the Kali VM configured on eth0 and the gateway at 10.0.0.1.

A ping scan was first run directly in Zenmap:

```
nmap -sn 10.0.0.0/24
```

The scan identified 2 live hosts on the subnet: 10.0.0.1 (the gateway, with MAC address 52:54:00:12:35:00, identified as a QEMU virtual NIC) and 10.0.0.2 (the Kali VM's own interface — nmap does not report a MAC address for the host's own local interface, which is expected/normal behaviour, not a finding).

To confirm the gateway's MAC address and capture the result to a log file, the ARP/neighbour cache was inspected and a second, ARP-based ping scan was run:

```
ip neigh
```

```
sudo nmap -sn -PR 10.0.0.0/24 | tee outputs/task6-mac-addresses.txt
```

The ip neigh output confirmed the gateway's MAC/ARP entry (10.0.0.1 → 52:54:00:12:35:00, marked STALE). The tee'd nmap -sn -PR scan reproduced the same two live hosts and the same gateway MAC address, saved to task6-mac-addresses.txt. (The output directory did not exist on the first attempt, so it was created with mkdir before the command was re-run successfully.)

Finally, the Topology tab in Zenmap was opened, the legend was enabled, and the resulting two-host topology (10.0.0.1 and 10.0.0.2 connected to the scanning host via dashed "no traceroute information" links) was saved as a PDF, consistent with the live hosts identified above.

5. Risk Analysis / Impact

The items below are reconnaissance-level observations from the footprinting and scanning activities, not confirmed vulnerabilities. No exploitation, credential attacks, privilege escalation or unauthorized access was attempted or performed as part of either module. The presence of a public email address, subdomain, IP address or MAC address does not by itself mean a system is vulnerable — further authorized testing would be required to confirm any actual weakness.

#	Finding / Observation	Evidence	Potential Impact	Risk Level
1	Public email addresses exposed	theHarvester (Baidu + all-sources) identified 2–3 microsoft.com email addresses via public sources.	Could assist an attacker in building target lists for phishing or social-engineering; reconnaissance value only.	● Medium
2	Large public subdomain / host footprint	theHarvester all-sources run identified 9,969 host/subdomain names and 153 associated IP addresses for microsoft.com from public sources.	A large public attack surface gives an attacker many potential entry points to research further; does not by itself indicate a vulnerability.	● Medium
3	Public ASN and infrastructure information	8 ASNs were identified for microsoft.com via public OSINT sources.	Helps an attacker map the target's hosting/network infrastructure for further authorized enumeration.	● Low
4	Live hosts discovered on local network	Zenmap/Nmap ping scan (nmap -sn 10.0.0.0/24) identified 2 live hosts: 10.0.0.1 and 10.0.0.2.	Confirms which devices are reachable on the LAN; unexpected/unknown devices should be investigated.	● Low
5	Gateway MAC address exposed via ARP	ip neigh and nmap -sn -PR identified the gateway (10.0.0.1) MAC address 52:54:00:12:35:00.	MAC/vendor information can help an attacker fingerprint network equipment; only useful from within the local segment.	● Low

Risk levels above are intentionally conservative: nothing found during these two OSINT/discovery modules rises above Medium, since no exploitation or credential exposure occurred.

6. Recommendations

- Review publicly exposed reconnaissance information — periodically check what email addresses, hostnames and ASN/infrastructure details about microsoft.com are discoverable via public OSINT tools.
- Review publicly exposed host/subdomain information — a very large number of subdomains (9,969) were surfaced through passive sources; organizations should periodically audit which subdomains are intentionally public.
- Maintain up-to-date DNS and network documentation so that discovered hosts and subdomains can quickly be matched against what is expected.
- Periodically perform authorized internal network discovery (e.g. Zenmap/Nmap ping scans) to confirm which devices are active on the local network.
- Investigate any unknown or unexpected devices identified during internal network discovery scans.
- Keep network topology documentation (such as the Zenmap topology diagram generated here) current as devices are added or removed.
- Perform all reconnaissance and scanning activity only within an authorized scope, as was done in this exercise.

7. Conclusion

During Week 2 of my Cybersecurity & Ethical Hacking internship at Networkwalks, I completed two practical modules: theHarvester-based footprinting of the microsoft.com domain (W2-PM4) and Zenmap-based network scanning of my own lab network (W2-PM5).

The footprinting exercise showed how much information about an organization can be gathered from entirely public, passive sources — including email addresses, thousands of subdomains, associated IP ranges and ASN ownership — without ever directly contacting the target's systems. The scanning exercise showed how a simple ping scan and ARP lookup can quickly identify live hosts and MAC addresses on a local network, and how Zenmap's topology view helps visualize those results.

Across both modules, I learned that reconnaissance and discovery are about building an accurate picture of a target, not about exploiting it — and that a good report should clearly separate what was observed from what is a confirmed security weakness. I also learned the practical importance of running tools only within an authorized scope: the footprinting module used only public OSINT sources against a third-party domain, while the scanning module was run only against a network I own.

8. Evidence Collected

The following screenshots and files were captured during the two exercises and are referenced throughout this report.

8.1 W2-PM4 — theHarvester Evidence

```
kali@kali: ~/Week2-PM4
Session Actions Edit View Help
kali@kali:~$ mkdir -p ~/Week2-PM4/screenshots,outputs
kali@kali:~$ cd ~/Week2-PM4
kali@kali:~/Week2-PM4$ theHarvester -d microsoft.com -l 1000 -b baidu | tee outputs/task1-theharvester-baidu.txt
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester *
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: microsoft.com

[*] Searching Baidu.

[*] No IPs found.

[*] Emails found: 2
is-microsoft-moreply@microsoft.com
microsoft-moreply@microsoft.com

[*] No people found.

[*] Hosts found: 10
adoption.microsoft.com
appsource.microsoft.com
community.fabric.microsoft.com
learn.microsoft.com
news.microsoft.com
office365.microsoft.com
service.microsoft.com
support.microsoft.com
watson.microsoft.com
wcpstatic.microsoft.com

kali@kali:~/Week2-PM4$
```

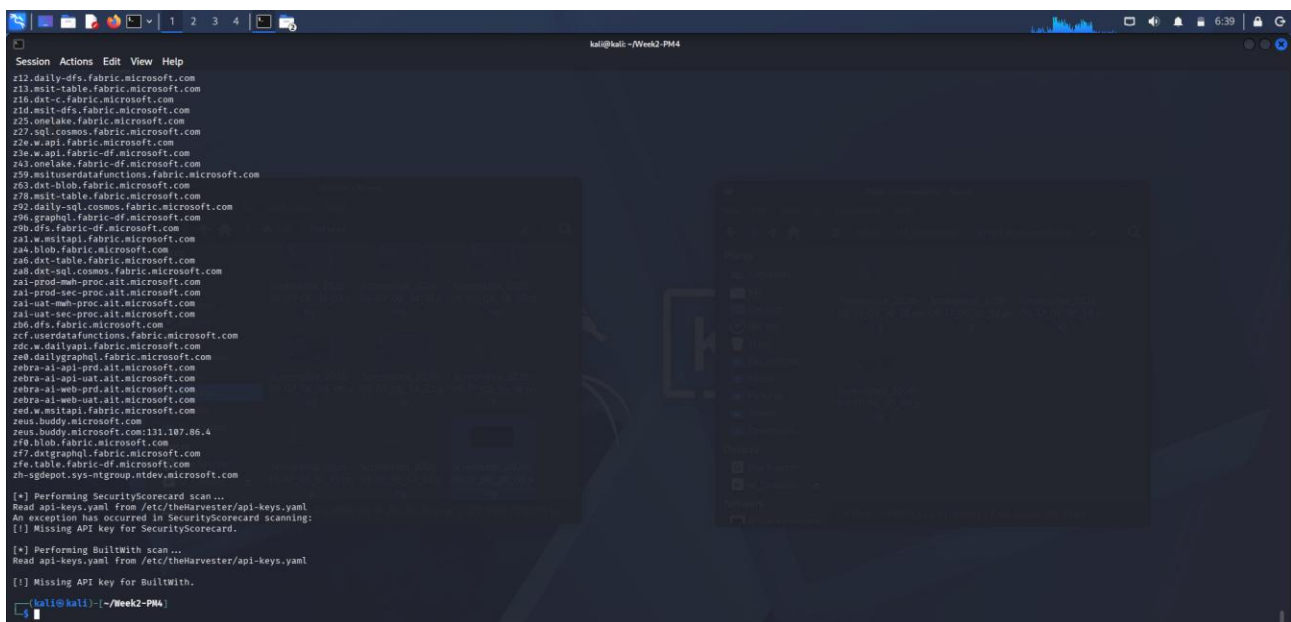
Task 1 — theHarvester -d microsoft.com -l 1000 -b baidu (Baidu source: 2 emails, 10 hosts).

```
kali@kali: ~/Week2-PM4
Session Actions Edit View Help
kali@kali:~/Week2-PM4$ theHarvester -d microsoft.com -l 50 -b all | tee outputs/task2-theharvester-all.txt
2026-09-17 06:33:18,358 - theHarvester.discovery.hudsonrocksearch - INFO - Starting Hudson Rock processing for: microsoft.com
2026-09-17 06:33:18,358 - theHarvester.discovery.hudsonrocksearch - INFO - Starting Hudson Rock search for: microsoft.com
2026-09-17 06:33:44,398 - theHarvester.discovery.hudsonrocksearch - INFO - Domain statistics: 602369 total compromised, 16140 employees, 581626 users
2026-09-17 06:33:44,400 - theHarvester.discovery.hudsonrocksearch - INFO - Extracted 43 hosts from employee UMLs
2026-09-17 06:33:45,401 - theHarvester.discovery.hudsonrocksearch - INFO - Hudson Rock search completed: found 43 hosts, 0 IPs, 0 emails
2026-09-17 06:33:45,401 - theHarvester.discovery.hudsonrocksearch - INFO - Hudson Rock processing completed successfully: 43 hosts, 0 IPs, 0 emails, 0 stealers
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
* theHarvester *
*
* theHarvester 4.10.1
* Coded by Christian Martorella
* Edge-Security Research
* cmartorella@edge-security.com
*
*****

[*] Target: microsoft.com

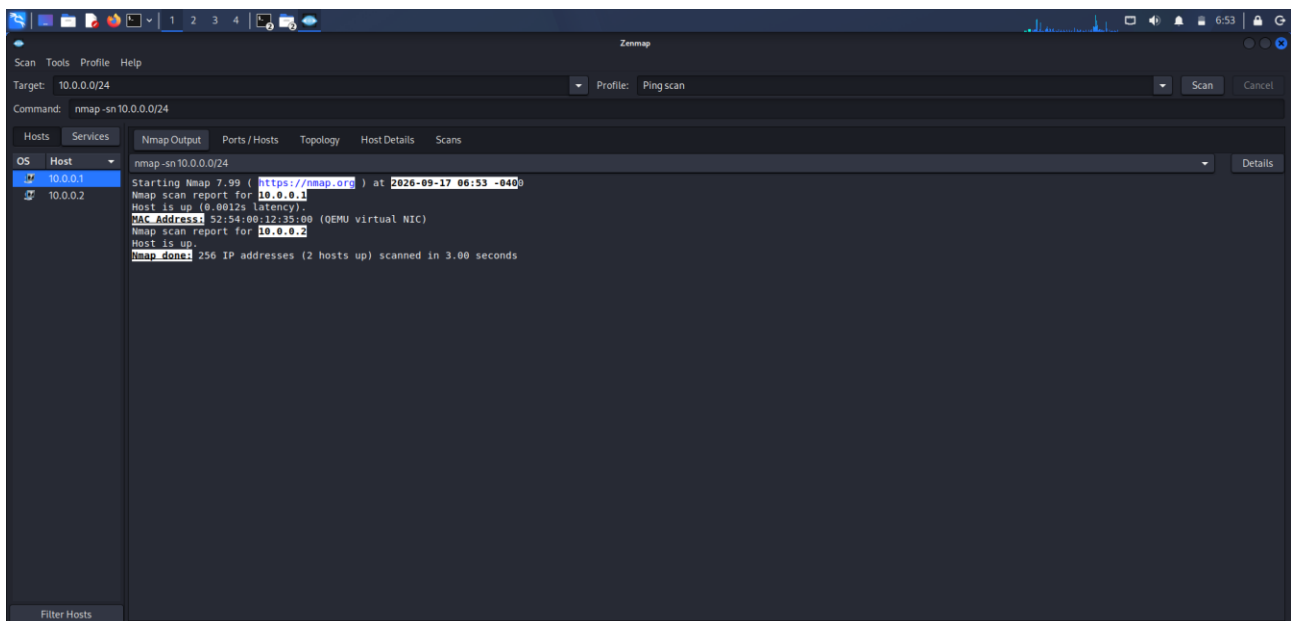
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bevigil search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bevigil.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
[!] Missing API key for Bitbucket.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process bufferoverrun search for word: 'microsoft.com'
Error Message:
[!] Missing API key for bufferoverrun.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to perform builtwith search for word: 'microsoft.com'
A Missing Key Error occurred in builtwith:
[!] Missing API key for Builtwith.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Failed to process brave search for word: 'microsoft.com'
Error Message:
[!] Missing API key for Brave Search.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Censys API key is missing or invalid:
[!] Missing API key for Censys ID and/or Secret.
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
Read api-keys.yaml from /etc/theHarvester/api-keys.yaml
```

Task 2 (start) — theHarvester -d microsoft.com -l 50 -b all: run initiated, most paid-API sources reporting missing keys as expected.



Task 2 (end) — theHarvester all-sources run in progress, showing subdomain results being returned from free/public sources.

8.2 W2-PM5 — Zenmap / Nmap Evidence



Ping scan — `nmap -sn 10.0.0.0/24` in Zenmap: 2 hosts up (10.0.0.1, 10.0.0.2).

```
kali@kali:~$ ip neigh
10.0.0.1 dev eth0 lladdr 52:54:00:12:35:00 STALE

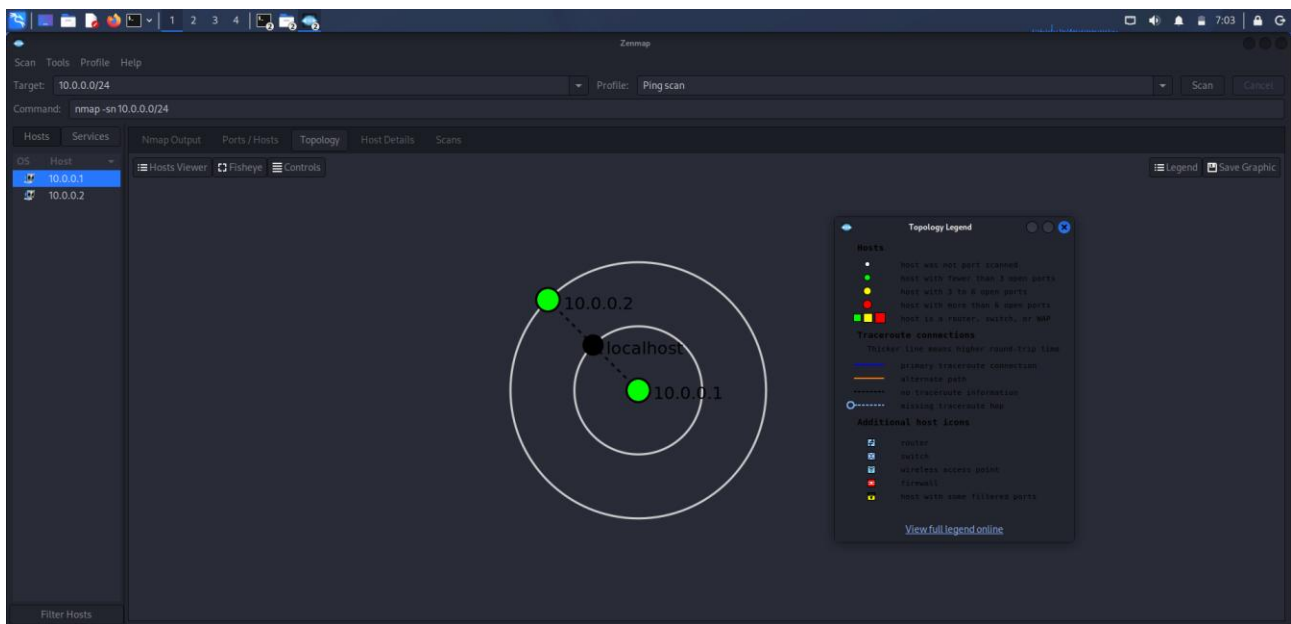
kali@kali:~$ sudo nmap -sn -PR 10.0.0.0/24 | tee ~/Week2-PM5/outputs/task6-mac-addresses.txt
tee: /home/kali/Week2-PM5/outputs/task6-mac-addresses.txt: No such file or directory
[sudo] password for kali:
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 06:57 -0400
Nmap scan report for 10.0.0.1
Host is up (0.001s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.0.2
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.94 seconds

kali@kali:~$ mkdir -p ~/Week2-PM5/[screenshots,outputs]

kali@kali:~$ sudo nmap -sn -PR 10.0.0.0/24 | tee ~/Week2-PM5/outputs/task6-mac-addresses.txt
Starting Nmap 7.99 ( https://nmap.org ) at 2026-09-17 06:59 -0400
Nmap scan report for 10.0.0.1
Host is up (0.00056s latency).
MAC Address: 52:54:00:12:35:00 (QEMU virtual NIC)
Nmap scan report for 10.0.0.2
Host is up.
Nmap done: 256 IP addresses (2 hosts up) scanned in 2.97 seconds

kali@kali:~$
```

MAC address confirmation — *ip neigh* and *sudo nmap -sn -PR 10.0.0.0/24* (tee'd to *task6-mac-addresses.txt*); gateway MAC 52:54:00:12:35:00.



Zenmap Topology view (legend enabled) — 10.0.0.1 and 10.0.0.2 shown relative to the scanning host.

Supporting file: PM5-Zenmap-Topology.pdf — the topology diagram exported from Zenmap, showing the same two hosts (10.0.0.1, 10.0.0.2) relative to the local scanning host, consistent with the ping scan results above.

9. Author / Project Information

Pentester Name (Cybersecurity Intern)	Karthik Raman Keerangudi Kalyanaraman
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